Kyle Wybo

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Professional Summary

Computer Science candidate with hands-on experience developing systems and embedded software in C, C++, and Python on Linux. Background spans real-time embedded systems, GPS timing and synchronization, object-oriented design, and full software development lifecycle practices in collaborative Agile environments. Strong foundation in systems programming, debugging, and technical documentation. U.S. citizen able to obtain a security clearance.

Technical Skills

Languages: C, C++, Python, Java, C#, SQL (PostgreSQL), JavaScript, HTML/CSS
Embedded & Systems: Linux administration, Bash scripting, networking/sockets, process & service management, GPS timing systems, Arduino, Raspberry Pi, Buildroot / embedded Linux
Software Engineering: Object-Oriented Design, Agile/Scrum, SDLC, Unit & Integration Testing, Debugging, Requirements Analysis, Design Documentation
Tools: Git, GitLab, Docker, Kubernetes, CI/CD pipelines, JIRA, Confluence, PostgreSQL, Google Cloud Platform

Education

Michigan Technological University — Houghton, MI
Bachelor of Science in Computer Science | GPA: 2.8 — *Aug. 2020 – April 2025*

Relevant Coursework: Embedded Systems, Systems Programming, Data Structures & Algorithms, Calculus I-II, Physics I-II

Experience

Undergraduate Research Assistant — Nov. 2024 – Aug. 2025
Michigan Technological University — *Houghton, MI*

- Developed real-time embedded software in C and Python for GPS clock synchronization and vehicle-to-infrastructure (V2X) communication systems, improving timing accuracy by 20% through iterative analysis and code refinement.
- Configured embedded Linux environments using Buildroot; administered Linux services, daemons, and networked processes across distributed hardware nodes.
- Wrote Bash scripts to automate system configuration, deployment, and environment setup across multiple Linux machines.
- Debugged socket-level networking failures and timing discrepancies between distributed embedded units; traced root causes using engineering documentation and systematic troubleshooting.
- Collaborated across a multi-disciplinary Agile team through all SDLC phases—requirements, design, implementation, integration, and test—producing design and test documentation throughout.

Projects

Intersection Traffic Observer | *React, Tailwind CSS, Docker, Git, SQL* — Jan. 2025 – April 2025

- Designed and built a real-time data-collection and processing pipeline applying object-oriented design and modular architecture to handle concurrent sensor data streams.
- Wrote unit and integration tests to verify correctness at each development phase; containerized the system with Docker for reproducible deployment.
- Integrated a PostgreSQL database for structured data storage; managed requirements traceability with JIRA and Confluence.

Gr4pp-L | *C#, Unity, Visual Studio, GitHub* — Sep. 2022 – Dec. 2023

- Designed and implemented modular, object-oriented software components in C# and Unity, maintaining clean architecture and reliable builds across a collaborative multi-developer team.
- Managed version control and collaborative workflows via GitHub; compiled program documentation and operating instructions throughout development.